4-Zone Conventional Fire Alarm Panel — Prototype Architecture & Power Tree

Scope: 4-zone conventional panel, 25-30 field devices, single-MCU architecture (no STM8, no shift registers), RS-485 present, talkback/GSM/multi-footprint OUT of scope for this prototype.
Basis: Ravel RE-2554/2558 reverse-engineering (Rev-Engg-Power-Circuit.xlsx), EMKIONYX BOM v1, existing hardware/electrical docs (mcu_selection.md, power_budget.md, system block diagrams).

SHEETS IN THIS WORKBOOK

Architecture	- block-level design, single-MCU GPIO budget, what changed vs the donor design
Quantity_BOM	- component quantities sized for 4 zones / 25-30 devices, single-MCU version
Power_Tree	- hierarchical Source -> Regulator -> Rail -> Load current/power model
Open_Decisions	- the 2 items that block a final power budget until you confirm them

HOW TO USE

Edit only the YELLOW cells in Power_Tree (per-load current assumptions) and Open_Decisions.
Every other cell is a formula that rolls child-load current up to its parent rail automatically.
Two scenarios are computed side by side: STANDBY and ALARM (worst case, per EN54-4 style budgeting, same method as the existing power_budget.md).

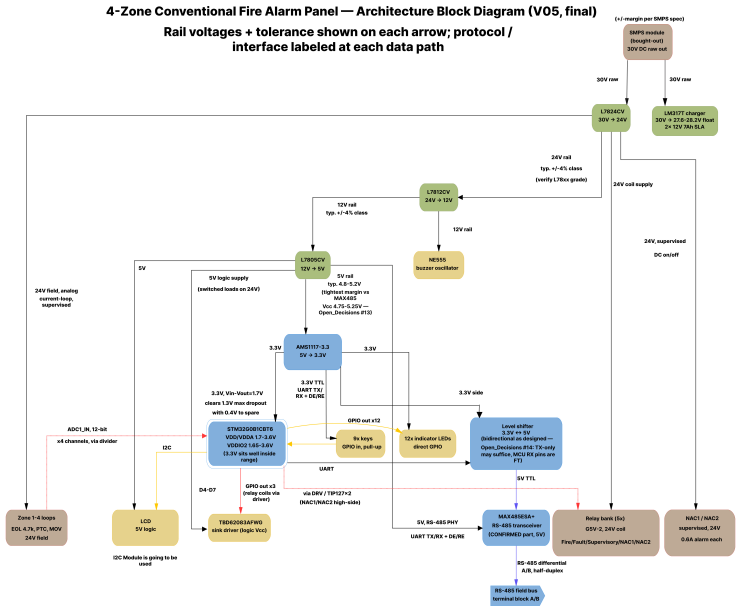
KEY CHANGES FROM THE REVERSE-ENGINEERED (Ravel-derived) DESIGN

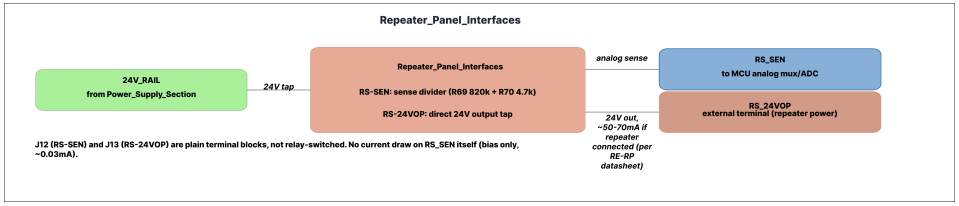
1. STM8S001J3 (display MCU) removed - STM32G0B1CBT6 alone drives ADC, RS-485, relays, NAC, and HMI.
2. 8x HEF4094BT shift registers removed - 12 LEDs + 9 keys + I2C LCD fit directly on STM32G0B1 GPIO.
3. R8C/25 (Renesas) removed entirely - was the donor's main MCU; STM32G0B1 replaces it too.
4. MAX485ESA+ (5V part) flagged for replacement with a 3.3V-native transceiver (SP3485/MAX3485/THVD1450) since STM32G0B1 is a 3.3V-only part (VDD/VDDA/VDDIO2 = 3.3V, no 5V-tolerant UART pins assumed here). Power_Tree is built assuming the 3.3V part with AMS1117.

CAUTION — REFERENCE DESIGNATORS ACROSS SOURCE DOCUMENTS

Component labels (Q1, U9, RL2, etc.) are NOT assumed to match between different source documents (the summary spreadsheet, the .kicad_sch, this workbook, any future upload). The same designator can refer to different physical parts in different documents, and the same physical part can carry different designators. Every cross-document claim in this workbook is traced by FUNCTION and net label (e.g. 'drives the NAC1 relay coil'), not by assuming 'Q1 in file A' = 'Q1 in file B'. When a conflict surfaces between sources (e.g. TIP127 vs BDX34C at the NAC relay driver position, resolved 2026 in favour of TIP127 per direct physical inspection of the board), direct inspection of the real hardware outranks any single document.

SYSTEM ARCHITECTURE





System Architecture — 4-Zone Conventional Panel, Single-MCU

1. SIGNAL FLOW (confirmed topology)

AC Mains -> SMPS module (bought-out, external) -> 30V DC raw output

30V raw -> L7824CV (linear reg) -> 24V regulated board rail (zones, relay coils, NAC, pre-reg chain)

30V raw -> LM317T (CC/CV charger, set 27.6-28.2V float) -> 2x 12V 7Ah SLA in series (24V battery string)

24V rail -> L7812CV (pre-regulator / thermal buffer) -> 12V rail (buzzer oscillator, feeds 5V stage)

12V rail -> L7805CV -> 5V rail (driver array logic, external LCD, MAX485 + level shifter)

5V rail -> AMS1117-3.3 LDO -> 3.3V rail (STM32G0B1CBT6, indicator LEDs, keypad, level-shifter 3.3V side)

STM32G0B1 <-> level shifter <-> MAX485ESA+ <-> RS-485 A/B terminal block (screw terminals, field bus)

STM32G0B1 GPIO -> TBD62083 (sink driver) -> 3x relay coils (Fire/Fault/Supervisory) on 24V

STM32G0B1 GPIO -> TIP127 PNP Darlington x2 (high-side) -> NAC1/NAC2 on 24V, EOL-supervised

STM32G0B1 ADC (8ch) <- voltage dividers <- Zone 1-4 loops, battery sense, mains/charger sense, rail health

STM32G0B1 GPIO -> Q_reset (NPN) -> shared zone-reset line -> drops 24V briefly to unlatch detectors

STM32G0B1 I2C -> external LCD module (bought-out)

STM32G0B1 GPIO (direct, no shift register) -> 12 indicator LEDs + 9 keys

2. WHY THIS DIFFERS FROM THE DONOR (RAVEL RE-2554/2558) BOM

Donor part/block	Removed / changed to	Reason
Renesas R5F21258SN (R8C/25, main MCU)	STM32G0B1CBT6	12-bit ADC vs 10-bit; dual watchdog + HW CRC for EN54-2 cl.13; active lifecycle
STM8S001J3M3TR (display/HMI MCU)	Removed — STM32G0B1 drives HMI directly	Only 1 MCU needed at 4-zone scale; removes inter-MCU UART link and second firmware image
8x HEF4094BT shift registers	Removed — direct GPIO	12 LEDs + 9 keys fit STM32G0B1's ~37 usable GPIO with margin; shift registers only earn their keep at 16-zone/32-LED scale
L7824CV (30V->24V)	KEPT — confirmed by you	30V raw SMPS output gives the LM317T charger headroom to hit 27.6-28.2V float; L7824CV makes the clean 24V board rail
MAX485ESA+ (5V RS-485 transceiver)	KEPT + level shifter added	Confirmed by you — level shifter (3.3V<->5V) bridges STM32G0B1's 3.3V UART to the 5V transceiver
TBD62083AFWG x2 (U7, U9)	Reduced to x1	Only 3 relay coils + 2 NAC-enable + buzzer-gate = 6 channels; one 8-channel driver covers it without the LED-matrix channels the donor needed

3. STM32G0B1CBT6 GPIO BUDGET (LQFP48, no shift register)

Function	Pins used	Notes
Zone ADC inputs (4 zones)	4	ADC1_IN, 12-bit, via divider network
Battery / mains / charger / rail-health sense	4	ADC1_IN, shared ADC peripheral
RS-485 UART + DE/RE	3	USART TX, RX, DE/RE via level shifter

Relay drive (Fire/Fault/Supervisory, via TBD62083)	3	GPIO out, sinks through driver
NAC1/NAC2 drive (via TIP127)	2	GPIO out
Zone reset	1	GPIO out -> Q_reset base
Buzzer gate (silence control)	1	GPIO out, gates NE555 output or reset
Indicator LEDs	12	Direct GPIO, current-limit resistor per LED
Keys (Silence, Menu, Enter, Evacuate, Lamp Test, cursor L/R, 2x spare/alphanumeric)	9	GPIO in, pull-up, debounce in firmware
LCD interface — JHD162A (parallel HD44780, 4-bit mode)	6	RS, E, D4-D7. NOTE: JHD162A is a native parallel-interface LCD, not I2C — 6 GPIO in 4-bit mode, not 2. Add a PCF8574 I2C backpack module if you want to free up 4 GPIO instead.
SWD (program/debug)	2	SWDIO, SWCLK — reserved, not app GPIO
Crystal/oscillator (if used)	2	OSC_IN/OUT — reserved if external crystal used
TOTAL used	49	of ~48 physical pins on LQFP48 - 49 pins are used so I2C drive LCD used.

Fits comfortably — no shift register or second MCU required at this scale. Confirm exact pin map at ICD stage (per README.md milestone 'ICD issued to firmware', day 8).

4. NEW FROM THE ACTUAL SCHEMATIC (.kicad_sch) — confirms/refines earlier assumptions

Donor board is the 8-zone variant, not 4-zone: connectors J8-J11 and J16-J19 are labelled zone 1-8. Confirms the 2x HEF4051BT analog mux design already noted in mcu_selection.md — zone count scales by populating more zone terminals + divider networks, not by adding MCU ADC pins. For our 4-zone prototype, populate 4 of the 8 positions.

Relay coil drivers are Q1/Q2 = BDX34C (PNP Darlington), not TIP127 as the earlier summary spreadsheet listed — corrected in Quantity_BOM.

Relay part is G5V-2, not HFD27/024-24V as earlier listed — coil voltage (5V vs 24V) is unconfirmed, see Open_Decisions.

RS-SEN (J12) and RS-24VOP (J13) are plain terminal blocks, not relay outputs — RS-24VOP taps 24V directly (likely repeater power), RS-SEN is a supervised sense input feeding the analog mux.

5. VOLTAGE MARGINS BY RAIL (regulator output tolerance, real datasheet figures)

Rail	Nominal	Regulator	Tolerance	Real-world range
30V raw	30V	SMPS module (bought-out)	Per SMPS datasheet — confirm with vendor	TBD — get from SMPS spec sheet
24V board rail	24V	L7824CV	+/-4% (standard 'C'-grade L78xx family)	23.04V - 24.96V
12V pre-reg	12V	L7812CV	+/-4%	11.52V - 12.48V

5V logic rail	5V	L7805CV	+/-4%	4.8V - 5.2V
3.3V MCU rail	3.3V	AMS1117-3.3	+/-1.5% typ (up to +/-2% depending on manufacturer grade)	3.235V - 3.365V (typ)
Battery float	27.6-28.2V	LM317T (adjustable CC/CV)	Set by feedback resistors, not a fixed IC tolerance	27.6V - 28.2V (design target)

AMS1117-3.3 dropout is ~1.1-1.3V typical — the 5V rail feeding it has ample headroom above 3.3V+dropout even at the low end of L7805CV's +/-4% range (4.8V still clears 3.3V+1.3V=4.6V comfortably).

6. COMMUNICATION PROTOCOLS BY INTERFACE

Interface	Protocol / signal type	Between	Notes
Zone sensing	Analog (ADC, 12-bit)	Zone dividers -> STM32G0B1 ADC	Not a digital protocol — a scaled analog voltage per zone
RS-485 field bus	RS-485 (half-duplex) over UART framing	STM32G0B1 UART -> level shifter -> MAX485ESA+ -> A/B terminal	MCU side is 3.3V UART logic; bus side is differential RS-485
LCD	Parallel, HD44780 4-bit mode	STM32G0B1 GPIO (RS, E, D4-D7) -> JHD162A	Not I2C — confirmed from JHD162A datasheet, corrected earlier in this build
HMI (LEDs, keys)	Direct GPIO, no protocol	STM32G0B1 GPIO <-> LEDs/keys	No shift register — direct-drive, confirmed feasible at this GPIO budget
Relay/NAC control	Direct GPIO -> driver stage	STM32G0B1 GPIO -> TBD62083/TIP127 -> relay coils	Logic-level control signal; coil current is separate, on 24V
Repeater link (RS-SEN, RS-24VOP)	Analog sense + direct 24V tap	Terminal blocks J12/J13 -> analog mux / 24V rail	Not a data protocol — confirmed from actual .kicad_sch as plain terminals, not relay-switched

Quantity BOM — sized for 4 zones / 25-30 field devices, single-MCU

STM32G0B1CBT6	LQFP48	1	3.3V	U1	Single MCU — see Architecture sheet
MAX485ESA+	SOIC-8	1	5V	U2	RS-485 transceiver, kept per your confirmation
Level shifter (3.3V<->5V, e.g. AMS1117 or TXB0104 or BSS138 4-ch module)	TSSOP14/module	1	3.3V/5V	U3	Bridges MCU UART+DE/RE to MAX485
L7824CV	TO-220	1	30V->24V	U4	30V raw -> 24V board rail
LM317T	TO-220	1	30V->28V	U5	CC/CV charger, 27.6-28.2V float, 0.8A max
L7812CV	TO-220	1	24V->12V	U6	Pre-regulator / thermal buffer
L7805CV	TO-220	1	12V->5V	U7	5V logic rail
3.3V LDO — AMS1117-3.3 (confirmed part)	SOT-223	1	5V->3.3V	U8	MCU + LED + keypad rail. Output tolerance ~+/-1.5% (up to +/-2% depending on manufacturer grade), dropout ~1.1-1.3V typ
TBD62083AFWG (8-ch sink driver)	SOP-24	1	5V logic / 24V switched	U9	3 relay coils + 2 NAC-enable + buzzer gate = 6 of 8 channels used
G5V-2 relay, 24V coil (Form C)	TO5/PCB relay	5	24V coil (confirmed)	K1-K5	Fire, Fault, Supervisory + NAC1, NAC2. Part number G5V-2 per actual .kicad_sch. Coil voltage CONFIRMED as 24V (your decision) — order the 24VDC-coil variant of G5V-2, not 5V/12V
Flyback diode (1N4148/1N4007)	DO-35/DO-41	5	24V	D1-D5	One per relay coil (5 relays now, was 3)

Quantity BOM — sized for 4 zones / 25-30 field devices, single-MCU

TIP127 (PNP Darlington, relay coil driver)	TO-220	2	24V coil drive	Q1, Q2	CONFIRMED by direct visual inspection of the physical Ravel board. The other engineer's .kicad_sch showed 'BDX34C' at this position — checked both datasheets: same TO-220 package, same Base-Collector-Emitter pinout, same 100V rating, both wildly overspecified for a relay coil's actual current, so no functional difference either way. Going with what you saw on the real hardware.
NE555	SOIC-8	1	12V	U10	MCU-independent fault buzzer oscillator (design decision D-005 — do not remove)
Piezo buzzer element	THT	1	12V (via NE555)	BZ1	Driven by NE555
NPN transistor (e.g. MMBT2222A)	SOT-23	1	24V switched	Q3	Shared zone-reset drive line
PPTC resettable fuse	THT/SMD	8	24V	F1-F8	1 per zone (4) + 1 per NAC (2) + battery input (1) + RS-485 bus (1)
MOV (varistor, 24V-rated)	THT/SMD	10	24V/field	MOV1-10	2 per zone (4x2=8) + 1 per NAC (2) — per donor topology
EOL resistor 4.7k ohm 1/4W	THT	6	Field	R_EOL1-6	1 per zone (4) + 1 per NAC (2); matches Ravel RE-RP spec (4.7K, 1/4W)
Zone divider resistor network (2x per zone)	THT/SMD	8	24V->ADC	R1-R8	Scales zone loop voltage below 3.3V ADC full-scale
LED, 3mm/5mm, various colours	THT	12	3.3V	LD1-LD12	4 zone Fire LEDs + System ON/AC ON/Battery ON/Charger ON/Fire/Fault/Supervisory/Silenced (8)
Tactile switch	THT/SMD	9	3.3V	SW1-SW9	Silence, Menu, Enter, Evacuate, Lamp Test, cursor L/R, 2x spare/alphanumeric
External LCD module (bought-out)	Module	1	5V (assumed)	LCD1	I2C character LCD; confirm exact part + current draw

Quantity BOM — sized for 4 zones / 25-30 field devices, single-MCU

2-pin terminal block, 5.08mm pitch	THT	9	Field/24V	J1-J9	Zone1-4 (4) + NAC1-2 (2) + Battery in (1) + SMPS/30V in (1) + spare (1)
2-pin terminal block (RS-485 A/B)	THT	1	Field	J10	RS-485 bus
2-pin terminal block (RS-SEN)	THT	1	24V/sense	J12	CONFIRMED from actual .kicad_sch — repeater/remote sense input, divider-fed to analog mux
2-pin terminal block (RS-24VOP)	THT	1	24V	J13	CONFIRMED from actual .kicad_sch — direct 24V output, likely for repeater panel power
3-pin terminal block (C/NO/NC)	THT	3	24V switched	J11-J13	Fire, Fault, Supervisory relay contacts
TVS diode (RS-485 bus protection)	SMD	2	Field	D4, D5	A/B line transient protection
Decoupling/bulk capacitors (assorted)	SMD/THT	35	All rails	C-misc	100nF per IC pin + bulk 10-470uF per regulator stage — itemize at schematic capture
2x 12V 7Ah SLA battery	External	2	24V string	BAT1, BAT2	Per power_budget.md — 24h standby + 30min alarm sizing

Note: field-device count (25-30 detectors/call-points) is set by your zone wiring, not by on-board BOM quantity — each zone loop is one 2-pole terminal block regardless of how many devices sit on it in para

Power Tree — Source -> Regulator -

Yellow cells = editable assumptions. Everything else is a for

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
3.3V RAIL (from AMS1117-3.3 LDO — confirme							
1	STM32G0B1CBT6 (Run mode, ADC+USART+GPIO active)	3.3	10	16	33	52.8	Datasheet-verified Table 28 base (~2-9mA depending on fHCLK/Range) + peripheral adders (Table 36) for ADC/USART/GPIO; alarm = higher activity, faster ADC scan, RS-485 traffic
1	Indicator LEDs (12x, direct GPIO, current-limit R)	3.3	24	72	79.2	237.6	Standby: 4 system LEDs lit (System/AC/Battery/Charger ON) @ ~6mA. Alarm: all 12 lit worst case @ ~6mA
1	Keypad pull-ups/pull-downs (9 keys)	3.3	1	1	3.3	3.3	Negligible, internal MCU pull-ups
1	Level shifter, 3.3V side	3.3	1	2	3.3	6.6	Logic-side draw only; 5V side counted on 5V rail
0	**3.3V RAIL TOTAL (LDO output)**	**3.3**	**36**	**91**	**118.8**	**300.3**	*Sum of all 3.3V loads above*
5V RAIL (from L7805CV)							
1	TBD62083AFWG driver, logic supply only	5	5	5	25	25	Logic Icc only — switched relay/NAC current flows on 24V, not through this rail
1	External LCD module — JHD162A (confirmed)	5	1.5	1.5	7.5	7.5	From JHD162A datasheet: +5V, 1.5mA. Replaces earlier generic-LCD assumption.
1	MAX485ESA+ RS-485 transceiver	5	2	4	10	20	Receiver-enabled idle ~0.3-1mA per datasheet; alarm scenario includes occasional TX bursts, engineering margin applied
1	Level shifter, 5V side	5	1	2	5	10	
1	(AMS1117) TXB0104PW-3.3 LDO input draw (= 3.3V rail current, reflected)	5	36	91	180	455	Linear LDO: Iin approx Iout (dropout small at these currents); formula-linked below
0	**5V RAIL TOTAL (L7805CV output)**	**5**	**45.5**	**103.5**	**227.5**	**517.5**	*Sum of all 5V loads above, including reflected 3.3V rail current through the LDO*
12V RAIL (from L7812CV — pre-regulator / the							
1	NE555 buzzer oscillator	12	3	15	36	180	Standby: quiescent bias only (silent). Alarm/fault: actively sounding, driving piezo
1	L7805CV input draw (= 5V rail current, reflected)	12	45.5	103.5	546	1242	

Power Tree — Source -> Regulator -

Yellow cells = editable assumptions. Everything else is a for

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
0	**12V RAIL TOTAL (L7812CV output)**	**12**	**48.5**	**118.5**	**582**	**1422**	*Sum of 12V loads, including reflected 5V rail current through L7805CV*

24V RAIL (from L7824CV — 30V raw -> 24V bo							
Field device / zone-loop assumptions (feeds							
	EOL resistor value (ohm)		4700				
	Devices per zone (avg, 25-30 total / 4 zones)		7				
	Device standby current (mA each) — 2-wire 24VDC detector spec		0.1				
	Device alarm current (mA, one device activated — loop is current-limited)		25				
1	Zone 1 loop (EOL + real detector spec)	24	5.806382979	30.106382979	139.353191489	722.553191489	Standby = EOL current + (devices x 0.1mA). Alarm = EOL current + 25mA (worst-case single device; the loop's 500-ohm current-limit divider caps combined multi-device alarm current well below a simple N x 25mA sum)
1	Zone 2 loop	24	5.806382979	30.106382979	139.353191489	722.553191489	Same formula as Zone 1
1	Zone 3 loop	24	5.806382979	30.106382979	139.353191489	722.553191489	Same formula as Zone 1
1	Zone 4 loop	24	5.806382979	30.106382979	139.353191489	722.553191489	Same formula as Zone 1
1	NAC1 (supervised, EOL 4.7k)	24	5	600	120	14400	Standby = supervision current only; alarm = full 0.6A sounder load
1	NAC2 (supervised, EOL 4.7k)	24	5	600	120	14400	Same basis as NAC1
1	Relay coil — RL3 Fire (de-energised in normal)	24	0	15	0	360	24V coil CONFIRMED (your decision). ~360mW coil, energises only in alarm
1	Relay coil — RL4 Fault (fail-safe: energised in normal)	24	15	15	360	360	24V coil CONFIRMED. Energised-in-normal is deliberate (dead panel must report itself as dead) — do not remove
1	Relay coil — RL5 Supervisory (energised in normal)	24	15	15	360	360	24V coil CONFIRMED. Also gates the buzzer per netlist (BUZ-1/BUZ-2 wired through RL5 contacts)
1	Relay coil — RL1 NAC1 (de-energised in normal)	24	0	15	0	360	24V coil CONFIRMED. Driven by Q1 TIP127 (confirmed by direct inspection of the physical board — see Quantity_BOM note)

Power Tree — Source -> Regulator -

Yellow cells = editable assumptions. Everything else is a for

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
1	Relay coil — RL2 NAC2 (de-energised in normal)	24	0	15	0	360	24V coil CONFIRMED. Driven by Q2 TIP127 (same confirmation as RL1)
1	Zone reset transistor drive	24	0	0	0	0	Pulsed only during reset command, negligible average current
1	RS-SEN sense divider (J12, R69 820k + R70 4.7k)	24	0.03	0.03	0.72	0.72	CONFIRMED from actual .kicad_sch: 2-pin terminal, divider feeds analog mux (U12/U13) — supervises repeater link presence/health. Bias current only, negligible
1	RS-24VOP output (J13, direct 24V tap)	24	0	0	0	0	CONFIRMED from actual .kicad_sch: 2-pin terminal, direct 24V output — no relay switching. 0mA in this prototype (no repeater connected yet). If a repeater is wired in later, add ~50-70mA per RE-RP datasheet (System Quiescent Current: 50-70mA)
1	L7812CV input draw (= 12V rail current, reflected)	24	48.5	118.5	1164	2844	
0	**24V RAIL TOTAL (L7824CV output)**	**24**	**111.755531915**	**1513.955531915**	**2682.132765957**	**36334.93276595**	*Sum of all 24V loads, including reflected 12V/5V/3.3V chain current through L7812CV*
1	L7824CV own quiescent current (own overhead, on 30V side)	30	5	5	150	150	Typical linear regulator IQ, drawn from the 30V input side

30V RAW (SMPS output, before L7824CV / LM317T)

1	L7824CV input draw (= 24V rail current, reflected)	30	111.755531915	1513.955531915	3352.665957447	45418.665957447	
1	L7824CV own quiescent current	30	5	5	150	150	
1	LM317T charger — quiescent (adjust pin + sense dividers)	30	8	8	240	240	Present continuously regardless of charge state
1	LM317T charger — active charge current	30	0	800	0	24000	0 in normal standby (battery topped up); up to 0.8A max during a recharge cycle, which is the PSU worst case, not the alarm case, per power_budget.md methodology
0	**30V RAW TOTAL DRAW (from SMPS)**	**30**	**124.755531915**	**2326.955531915**	**3742.665957447**	**69808.665957447**	*This is the current the SMPS module must actually deliver*

SMPS SIZING — YOUR OWN CALCULATION (

Power Tree — Source -> Regulator -

Yellow cells = editable assumptions. Everything else is a for

	Node / Load	Voltage (V)	I Standby (mA)	I Alarm (mA)	P Standby (mW)	P Alarm (mW)	Notes
	Design margin over worst-case draw		0.25	*(edit — 25% shown, matches the derating style already used in power_budget.md)*			
	Worst case = ALARM + active battery recharge simultaneously		**124.755531915**	**2326.955531915**			
	Recommended SMPS continuous rating (mA), with margin			**2908.694414894**			
	Recommended SMPS rating (A), with margin			**2.91**			

Buy an SMPS module rated 30V DC output at or above the computed Amps figure above. This is OUR number, derived bottom-up from this panel's actual loads — not Ravel's 2.5A

BATTERY SIZING (24V string) — same metho

	Standby current on 24V-equivalent bus (mA)		111.755531915
	Alarm current on 24V-equivalent bus (mA)		1513.955531915
	Standby duration required (h)		24
	Alarm duration required (h)		0.5
	Derating factor (ageing/temp/EOD)		1.25
	Required battery capacity (Ah)		**4.30**

Compare against selected battery (e.g. 2x 12V 7Ah = 7Ah p

Open Decisions / Assumptions Log

#	Item	Current assumption in this workbook	Status
1	SMPS module output voltage/current	30V DC raw (per your confirmation, headroom for 28.2V float charging). Current rating is CALCULATED in Power_Tree, not assumed — see 'Recommended SMPS rating' near the bottom of that sheet.	RESOLVED
2	RS-485 transceiver	MAX485ESA+ (5V) kept, with a 3.3V<->5V level shifter added between it and STM32G0B1's UART.	RESOLVED
3	Shift registers / second MCU	Removed. Single STM32G0B1CBT6 drives HMI directly via GPIO — see Architecture sheet GPIO budget.	RESOLVED
4	Field detector current draw (25-30 devices)	RESOLVED with real spec: 2-wire/24VDC, standby <=0.1mA, alarm 25mA. Zone loop current is now a live formula (EOL resistor + device count) in Power_Tree, not a copied benchmark.	RESOLVED
5	External LCD module exact part	RESOLVED: JHD162A, +5V, 1.5mA per datasheet.	RESOLVED
6	LCD interface type	LCD supporting I2C pins are used. No external backpack is needed.	RESOLVED
7	RL1/RL2 — NAC1/NAC2 relays	RESOLVED: confirmed present in the reverse-engineered netlist (RL1 -> NAC1, RL2 -> NAC2), same HFD27/024 spec as Fire/Fault/Supervisory. Added to Quantity_BOM (relay qty 3->5) and Power_Tree (24V rail).	RESOLVED
8	RS-SEN and RS-24VOP	RESOLVED from the actual .kicad_sch: both are plain 2-pin terminal blocks (J12, J13), not relay outputs. RS-24VOP = direct 24V output tap (likely repeater panel power). RS-SEN = supervised sense input (R69 820k/R70 4.7k divider into the analog mux U12/U13), monitors a remote link's presence/health. Added to Quantity_BOM and Power_Tree.	RESOLVED

9	Battery capacity vs updated load model	Power_Tree recomputes required Ah live from the current load model. Relay coil voltage (item 11) is now resolved at 24V, so this figure is stable — recheck again only if detector count/spec or LCD assumptions change further.	RESOLVED — recheck if load assumptions change again
10	LM317T charger headroom at 30V input	27.6-28.2V float target from 30V raw leaves only ~1.8-2.4V headroom. LM317 typical dropout is ~2-3V at rated current — fine at the charger's modest 0.8A max, but confirm against the specific LM317T grade/heatsinking you use.	**OPEN — verify at bring-up (matches PB-3 in power_budget.md verification table)**
11	Relay coil voltage — K1-K5 (G5V-2)	RESOLVED: 24V coil, per your decision. Rationale: fail-safe relays (Fault/Supervisory) should draw power from the rail closest to the battery, not through 2 extra regulation stages; less wasted heat at 24V than 5V for the same coil power; keeps the already-busier 5V rail lighter. Order the 24VDC-coil variant of G5V-2.	RESOLVED
12	TIP127 vs BDX34C relay driver	RESOLVED in favour of TIP127, per your direct visual inspection of the physical Ravel board — this outranks the other engineer's .kicad_sch, which showed BDX34C at this position (likely a transcription slip or a substitution on their end, not confirmed). Checked both datasheets: same TO-220 package, same pinout, same 100V rating, both far exceed what a relay coil actually draws — so even if BDX34C is what's really there, no functional consequence. Quantity_BOM reverted to TIP127.	RESOLVED